THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 4
2026/02/11 and 2026/02/13

1. Show that if f and g are two density functions and $0 \leq \lambda \leq 1$, then $\lambda f + (1-\lambda)g$ is also a density function. Is the product fg still a density function?

   Solution:

   Since $f(x)$ and $g(x)$ are nonnegative for any $x \in \mathbb{R}$, we have $\lambda f(x) + (1-\lambda)g(x) \geq 0$ for any $x \in \mathbb{R}$ and $0 \leq \lambda \leq 1$. Moreover, the function $\lambda f + (1-\lambda)g$: $\mathbb{R} \to [0, \infty)$ is integrable with

   $$\int_{-\infty}^{\infty} \lambda f(x) + (1-\lambda)g(x)dx = \lambda \int_{-\infty}^{\infty} f(x)dx + (1-\lambda)\int_{-\infty}^{\infty} g(x)dx = \lambda + (1-\lambda) = 1.$$

   Hence, $\lambda f + (1-\lambda)g$ is still a density function. For the product fg, we may consider a counterexample:

   $$f(x) = \begin{cases} x & 0 \leq x \leq \sqrt{2}, \\ 0 & \text{otherwise} \end{cases}$$

   and let $g(x) = f(x)$. Then $\int f dx = \int g dx = 1$, but

   $$\int_{-\infty}^{\infty} f(x)g(x)dx = \int_0^{\sqrt{2}} x^2 dx = \frac{2\sqrt{2}}{3} \neq 1.$$

2. Find the expected value and variance of the random variable X when

   (a) $X \sim Geom(p)$;
   (b) $X \sim NB(r, p)$;
   (c) $X \sim Poisson(\lambda)$;
   (d) $X \sim U[a, b]$;
   (e) $X \sim Exp(\lambda)$;
   (f) $X \sim Gamma(\alpha, \lambda)$;

   Solution:

   (a) $X \sim Geom(p)$; The p.m.f. of X is $P(X = k) = (1-p)^{k-1}p$. So

   $$E[X] = \sum_{k=1}^{\infty} kP(X = k) = \sum_{k=1}^{\infty} k(1-p)^{k-1}p.$$

To evaluate this series, one can separate each k into sum of 1 s. i.e.

$$\begin{aligned}
& \sum_{k=1}^{\infty} k(1-p)^{k-1}p \\
= \; & p + (1-p)p + (1-p)p + (1-p)^2p + (1-p)^2p + (1-p)^2p + \cdots \\
= \; & (p + (1-p)p + (1-p)^2p + \cdots) + ((1-p)p + (1-p)^2p + \cdots) + ((1-p)^2p + \cdots) + \cdots \\
= \; & \frac{p}{1-(1-p)} + \frac{(1-p)p}{1-(1-p)} + \frac{(1-p)^2p}{1-(1-p)} + \cdots \\
= \; & 1 + (1-p) + (1-p)^2 + \cdots \\
= \; & \frac{1}{p}.
\end{aligned}$$

For $E(X^2)$, one can use the similar technique by the fact that $n^2 = 1 + 3 + \cdots + (2n-1)$ for $n \in \mathbb{N}$.

$$\begin{aligned}
& E[X^2] \\
= \; & \sum_{k=1}^{\infty} k^2(1-p)^{k-1}p \\
= \; & p + (1-p)p + 3(1-p)p + (1-p)^2p + 3(1-p)^2p + 5(1-p)^2p + \cdots \\
= \; & (p + (1-p)p + (1-p)^2p + \cdots) + 3((1-p)p + (1-p)^2p + \cdots) + 5((1-p)^2p + \cdots) + \cdots \\
= \; & \frac{p}{1-(1-p)} + \frac{3(1-p)p}{1-(1-p)} + \frac{5(1-p)^2p}{1-(1-p)} + \cdots \\
= \; & \sum_{k=1}^{\infty}(2k-1)(1-p)^{k-1} \\
= \; & \frac{2}{p}\sum_{k=1}^{\infty} k(1-p)^{k-1}p - \sum_{k=1}^{\infty}(1-p)^{k-1} \\
= \; & \frac{2}{p} \times \frac{1}{p} - \frac{1}{p} = \frac{2-p}{p^2}.
\end{aligned}$$

Therefore $Var(X) = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}$.

(b) $X \sim NB(r,p)$; The p.m.f. of X is $P(X=k) = \mathbf{C}_{r-1}^{k-1}(1-p)^{k-r}p^r$. One can verify that this is indeed a p.m.f. by checking the sum of all possible k add up to 1:

$$\sum_{k=r}^{\infty} P(X=k) = \sum_{k=r}^{\infty} \mathbf{C}_{r-1}^{k-1}(1-p)^{k-r}p^r = \sum_{l=0}^{\infty} \mathbf{C}_{l}^{r-1+l}(1-p)^{l}p^r$$

where we let $l = k - r$ and use the fact that $\mathbf{C}_r^n = \mathbf{C}_{n-r}^n$.

Let $S_n = \sum_{l=0}^{\infty} \mathbf{C}_l^{n+l}(1-p)^l$. By the fact that $\mathbf{C}_r^n = \mathbf{C}_{r-1}^{n-1} + \mathbf{C}_r^{n-1}$,

$$\begin{aligned} S_n = & \ \mathbf{C}_0^n(1-p)^0 + \sum_{l=1}^{\infty}(\mathbf{C}_{l-1}^{n+l-1} + \mathbf{C}_l^{n-1+l})(1-p)^l \\ = & \ (1-p)\sum_{l=1}^{\infty}\mathbf{C}_{l-1}^{n+l-1}(1-p)^{l-1} + \sum_{l=1}^{\infty}\mathbf{C}_l^{n-1+l}(1-p)^l + 1 \\ = & \ (1-p)\sum_{l'=0}^{\infty}\mathbf{C}_{l'}^{n+l'}(1-p)^{l'} + \sum_{l=0}^{\infty}\mathbf{C}_l^{n-1+l}(1-p)^l \\ = & \ (1-p)S_n + S_{n-1} \end{aligned}$$

where we let $l' = l-1$. So $pS_n = S_{n-1} \Rightarrow S_n = \frac{1}{p}S_{n-1} = \cdots = \frac{1}{p^n}S_0$. Since $S_0 = \sum_{l=0}^{\infty}\mathbf{C}_l^l(1-p)^l = \frac{1}{p}$, we have $S_n = \frac{1}{p^{n+1}}$. Therefore

$$\sum_{k=r}^{\infty}P(X=k) = \sum_{l=0}^{\infty}\mathbf{C}_l^{r-1+l}(1-p)^l p^r = p^r S_{r-1} = \frac{p^r}{p^{r-1+1}} = 1.$$

Now we proceed to $E[X]$.

$$\begin{aligned} E(X) = & \ \sum_{k=r}^{\infty}k\mathbf{C}_{r-1}^{k-1}(1-p)^{k-r}p^r \\ = & \ \sum_{k=r}^{\infty}k\frac{(k-1)!}{(r-1)!(k-r)!}(1-p)^{k-r}p^r \\ = & \ \frac{r}{p}\sum_{k=r}^{\infty}\frac{k!}{r!(k-r)!}(1-p)^{k-r}p^{r+1} \\ = & \ \frac{r}{p}\sum_{k'=r'}^{\infty}\frac{(k'-1)!}{(r'-1)!(k'-r')!}(1-p)^{k'-r'}p^{r'} \\ = & \ \frac{r}{p} \end{aligned}$$

where we let $k' = k+1$ and $r' = r+1$.
Similarly

$$\begin{aligned} E(X^2) = & \ \sum_{k=r}^{\infty}k^2\mathbf{C}_{r-1}^{k-1}(1-p)^{k-r}p^r \\ = & \ \frac{r}{p}\sum_{k=r}^{\infty}(k+1-1)\frac{k!}{r!(k-r)!}(1-p)^{k-r}p^{r+1} \\ = & \ \frac{r(r+1)}{p^2}\sum_{k=r}^{\infty}\frac{(k+1)!}{(r+1)!(k-r)!}(1-p)^{k-r}p^{r+2} - \frac{r}{p}\sum_{k=r}^{\infty}\frac{k!}{r!(k-r)!}(1-p)^{k-r}p^{r+1} \\ = & \ \frac{r(r+1)}{p^2} - \frac{r}{p} = \frac{r^2+r-rp}{p^2} \end{aligned}$$

Therefore $Var(X) = \frac{r^2+r-rp}{p^2} - \frac{r^2}{p^2} = \frac{r(1-p)}{p^2}$.

Remark: A more convenient way to compute $E(X)$ and $Var(X)$ is to use the fact that X can be represented as the sum of r independent geometric random variables with parameter p.

(c) $X \sim Poisson(\lambda)$: The p.m.f. of X is $P(X = k) = \dfrac{\lambda^k}{k!}e^{-\lambda}$.

We can verify that it is indeed a p.m.f. by the fact that the Taylor's expansion of $e^{\lambda x}$ with respect to $x = 0$ is $\displaystyle\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}x^k$.

So by putting $x = 1$ we have $\displaystyle\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$ and thus $\displaystyle\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}e^{-\lambda} = 1$. Now

$$E(X) = \sum_{k=0}^{\infty} k\frac{\lambda^k}{k!}e^{-\lambda} = \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!}e^{-\lambda} = \sum_{k=1}^{\infty} \lambda\frac{\lambda^{k-1}}{(k-1)!}e^{-\lambda} = \lambda\sum_{k'=0}^{\infty} \frac{\lambda^{k'}}{(k')!}e^{-\lambda} = \lambda$$

where we let $k' = k - 1$ at the end. Similarly

$$E(X^2) = \sum_{k=0}^{\infty} k^2\frac{\lambda^k}{k!}e^{-\lambda} = \sum_{k=1}^{\infty}(k-1+1)\frac{\lambda^k e^{-\lambda}}{(k-1)!} = \lambda^2\sum_{k=2}^{\infty} \frac{\lambda^{k-2}e^{-\lambda}}{(k-2)!} + \lambda\sum_{k=1}^{\infty} \frac{\lambda^{k-1}e^{-\lambda}}{(k-1)!} = \lambda^2+\lambda.$$

Therefore $Var(X) = E(X^2) - E(X)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$.

(d) $X \sim U[a, b]$: The p.d.f. of X is $p(x) = \frac{1}{b-a}$.

$$E(X) = \int_a^b \frac{x\,dx}{b-a} = \left[\frac{x^2}{2(b-a)}\right]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{b+a}{2}.$$

$$E(X^2) = \int_a^b \frac{x^2\,dx}{b-a} = \left[\frac{x^3}{3(b-a)}\right]_a^b = \frac{b^3-a^3}{3(b-a)} = \frac{b^2+ab+a^2}{3}$$

by the fact that $b^n - a^n = (b-a)(b^{n-1} + b^{n-2}a + \cdots + ba^{n-2} + a^{n-1})$.

So $Var(X) = \dfrac{b^2+ab+a^2}{3} - \dfrac{(b+a)^2}{4} = \dfrac{4b^2+4ab+4a^2-3a^2-6ab-3b^2}{12} = \dfrac{(b-a)^2}{12}$.

(e) $X \sim Exp(\lambda)$: The p.d.f. of X is $p(x) = \lambda e^{-\lambda x}$.

$$E(X) = \int_0^{\infty} x\lambda e^{-\lambda x}dx = -\int_0^{\infty} x de^{-\lambda x} = -\left[xe^{-\lambda x}\right]_0^{\infty} + \int_0^{\infty} e^{-\lambda x}dx = \frac{-1}{\lambda}\int_0^{\infty} de^{-\lambda x} = \frac{1}{\lambda}$$

where $\lim\limits_{x\to\infty} xe^{-\lambda x} = 0$ by L'Hôspital's rule. Similarly

$$\begin{aligned} E(X^2) &= \int_0^{\infty} x^2\lambda e^{-\lambda x}dx \\ &= -\int_0^{\infty} x^2 de^{-\lambda x} \\ &= -\left[x^2e^{-\lambda x}\right]_0^{\infty} + \frac{2}{\lambda}\int_0^{\infty} x\lambda e^{-\lambda x}dx \\ &= 0 + \frac{2}{\lambda}\left(\frac{1}{\lambda}\right) = \frac{2}{\lambda^2} \end{aligned}$$

where again $\lim\limits_{x\to\infty} x^2e^{-\lambda x} = 0$ by L'Hôspital's rule and we use the result of $E(X)$.

Therefore $Var(X) = E(X^2) - E(X)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$.

Remark: These can also be obtained by the results of the Gamma distribution, since $Exp(\lambda)$ is a special case of $Gamma(\alpha, \lambda)$ with $\alpha = 1$.

(f) $X \sim Gamma(\alpha, \lambda)$: The p.d.f. of X is $p(x) = \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}$. We have

$$\int_0^\infty p(x)dx = \frac{\int_0^\infty \lambda e^{-\lambda x}(\lambda x)^{\alpha-1}dx}{\Gamma(\alpha)} = \frac{\int_0^\infty e^{-\lambda x}(\lambda x)^{\alpha-1}d(\lambda x)}{\Gamma(\alpha)} = \frac{\Gamma(\alpha)}{\Gamma(\alpha)} = 1$$

since $\Gamma(\alpha) = \int_0^\infty e^{-x}x^{\alpha-1}dx$. So

$$\begin{aligned} E(X) = & \frac{\int_0^\infty \lambda x e^{-\lambda x}(\lambda x)^{\alpha-1}dx}{\Gamma(\alpha)} \\ = & \frac{\int_0^\infty e^{-\lambda x}(\lambda x)^{\alpha}d(\lambda x)}{\lambda\Gamma(\alpha)} \\ = & \frac{\Gamma(\alpha+1)}{\lambda\Gamma(\alpha)} \\ = & \frac{\alpha\Gamma(\alpha)}{\lambda\Gamma(\alpha)} = \frac{\alpha}{\lambda} \end{aligned}$$

since $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$. Similarly,

$$\begin{aligned} E(X^2) = & \frac{\int_0^\infty \lambda x^2 e^{-\lambda x}(\lambda x)^{\alpha-1}dx}{\Gamma(\alpha)} \\ = & \frac{\int_0^\infty e^{-\lambda x}(\lambda x)^{\alpha+1}d(\lambda x)}{\lambda^2\Gamma(\alpha)} \\ = & \frac{\Gamma(\alpha+2)}{\lambda^2\Gamma(\alpha)} \\ = & \frac{(\alpha)(\alpha+1)\Gamma(\alpha)}{\lambda^2\Gamma(\alpha)} = \frac{\alpha^2+\alpha}{\lambda^2}. \end{aligned}$$

Therefore $Var(X) = E(X^2) - E(X)^2 = \frac{\alpha^2+\alpha}{\lambda^2} - \frac{\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}$.

3. Prove the following theorem for inverse transform sampling: For any random variable X with c.d.f F and $U \sim U[0,1]$, the random variable $F^{-1}(U)$ has the same distribution as X, where $F^{-1} : [0,1] \to \mathbb{R}$ is defined as

$$F^{-1}(u) = \inf\{x \mid F(x) \geq u\}.$$

Solution:

Let $Y = F^{-1}(U)$ and $F_Y(x) := P(Y \leq x)$ be the c.d.f. of Y. It suffices to show that $F_Y(x) = F(x)$ for any $x \in \mathbb{R}$. Note that $F_Y(x) = P(F^{-1}(U) \leq x)$ and $F(x) = P(U \leq F(x))$, hence it suffices to show that $P(F^{-1}(U) \leq x) = P(U \leq F(x))$ for any $x \in \mathbb{R}$. We claim that $F^{-1}(u) \leq x$ if and only if $u \leq F(x)$ for any $x \in \mathbb{R}$ and $u \in [0,1]$.

- ($\Rightarrow$): If $F^{-1}(u) \leq x$, then by the monotonicity of F we have $F(F^{-1}(u)) \leq F(x)$. Since F is right-continuous, we have $F(F^{-1}(u)) \geq u$ (to see this, let $x^* = F^{-1}(u) = \inf\{x \mid F(x) \geq u\}$, then there exists a decreasing sequence $x_k \to x^*$ with $F(x_k) \geq u$, which implies $F(x^*) \geq u$ by the right-continuity of F). Hence $u \leq F(x)$.
- ($\Leftarrow$): If $u \leq F(x)$, then $x \in \{x' \mid F(x') \geq u\}$. Hence $F^{-1}(u) = \inf\{x' \mid F(x') \geq u\} \leq x$.

Thus we have

$$P(F^{-1}(U) \leq x) = P(U \in \{u \mid F^{-1}(u) \leq x\}) = P(U \in \{u \mid u \leq F(x)\}) = P(U \leq F(x)),$$

which completes the proof.